

Next Generation Data Centers and Energy Management Systems

Lecture 6A: Chip Cooling: Heat Conduction and Thermal Interface Materials

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Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- Liquid Cooling & Advanced Heat Transfer
- **Chip Cooling: Heat Conduction and Thermal Interface Materials**
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Understand the thermodynamic limits of computing (Landauer's Principle).
- Analyze modern chip architectures and their thermal implications.
- Model heat conduction through a multi-layer chip cooling stack.

Typical layout of a data center facility

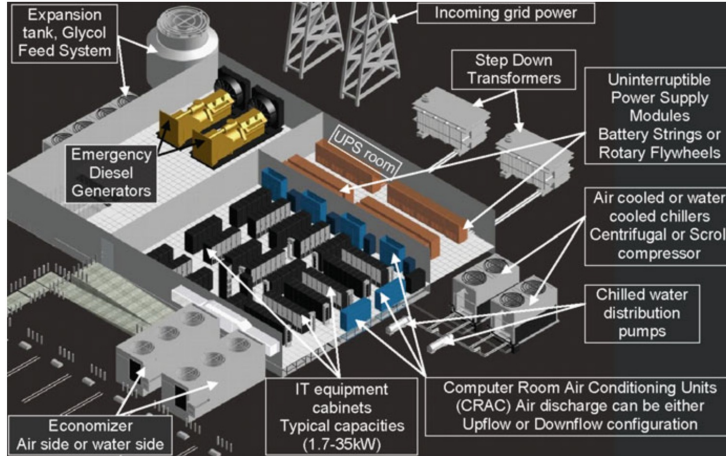


Figure 1: (Figure 1.1 in Ref. [1]).

Project 1: Feedback

- Check that you have addressed every task listed in the project tasks section of the description before submitting.
- Use the template. Its purpose is to simplify your lives. You don't need to add sections, modify the text size, etc.
- Ben: Come to office hours or send me questions via email.

Seminar 3

- Michael Donahue, Schneider Electric
- Title: TBD
- Date: October 8, 2026
- Time: 15:30 (Berkeley Time)
- Location: GSPP 150

Thermodynamics of Computing

- **Why is heat generated?** Irreversible computing operations (e.g., erasing a bit) increase entropy.
- **Landauer's Principle:** The minimum energy required to erase one bit of information is $E_{min} = k_B T \ln 2$.
- **Metric:** Joules per FLOP (or bit flip).
- **Modern vs. Limit:** Modern systems are orders of magnitude away from the thermodynamic limit.

Modern Chip Architecture

- **Components:**

- Logic Die (CPU/GPU cores)
- Memory (HBM - High Bandwidth Memory)
- Interposer (Silicon or organic substrate connecting die and memory)
- Vias (TSVs - Through-Silicon Vias)

- **Heat Generation:** Concentrated in the logic die (high power density).

- **Thermal Management Stack:**

- Thermal Interface Material (TIM)
- Heat Spreader (Integrated Heat Spreader - IHS)
- Vapor Chamber (Phase change cooling)
- Heat Sink (Convection to air/liquid)

Heat Transfer Modeling: Convection

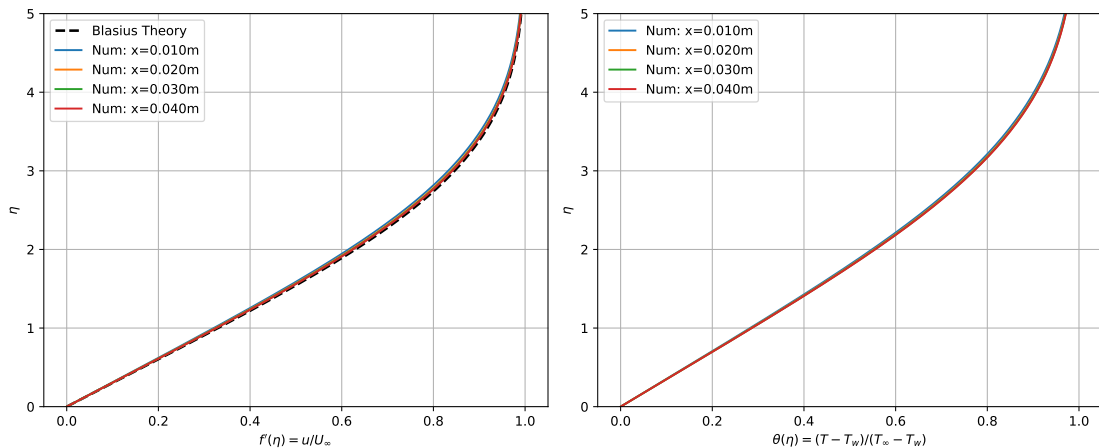


Figure 2: Exact similarity solution vs. numerical solution.

Heat Transfer Modeling: Convection

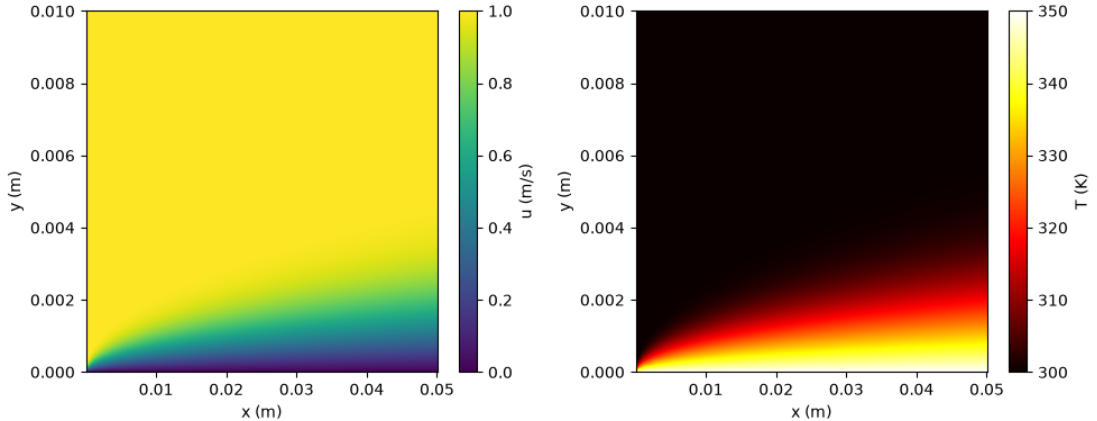


Figure 3: Exact similarity solution.

Heat Transfer Modeling: Convection

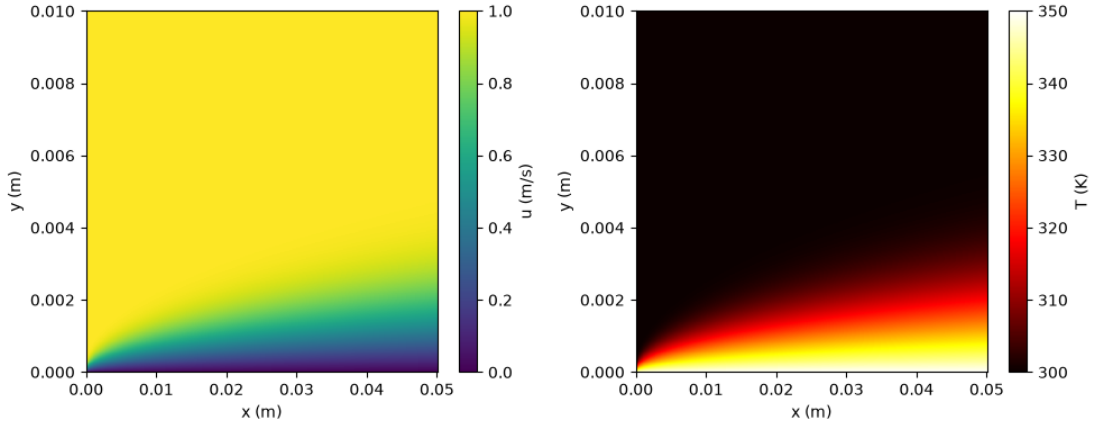


Figure 4: Numerical solution.

Heat Transfer Modeling: Conduction

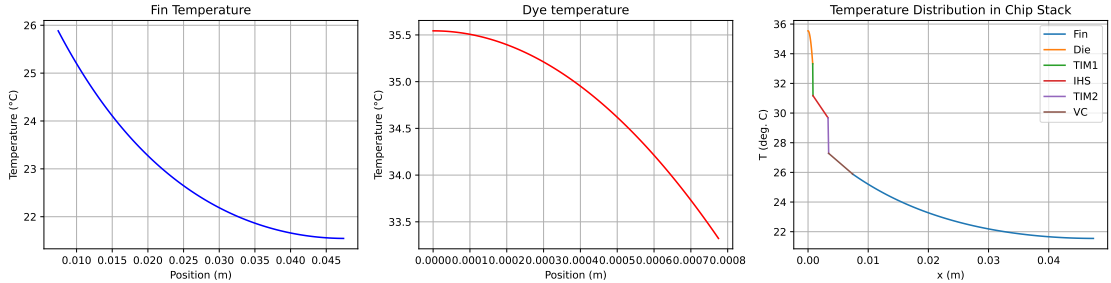


Figure 5: Temperature distribution in the chip stack.

References I

- [1] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.